Think of one of the grandest and most obvious phenomena: the diurnal rotation of the skies. It is the most immediate and magnificent characteristic of the universe around us: it turns. But is this turning really a characteristic of the universe? It is not. It took us thousands of years, but in the end we managed to understand the revolving of the heavens: we understood that it is *we* who turn, not the universe. The rotation of the heavens is a perspective effect due to our particular way of moving on Earth, rather than a mysterious property of the dynamics of the universe.

Something similar might be true for time's arrow. The low initial entropy of the universe might be due to the particular way in which we—the physical system that we are part of—interact with it. We are attuned to a very particular subset of aspects of the universe, and it is *this* that is oriented in time.

How can a particular interaction between us and the rest of the world determine a low initial entropy?

It's simple. Take a pack of twelve cards, six red and six black. Arrange it so that the red cards are all at the front. Shuffle the pack a little and then look for the black cards that have ended up among the red ones. Before shuffling, there are none; after, some. This is a basic example of the growth of entropy. At the start of the game, the number of black cards among the red in the first half of the pack is zero (the entropy is low) because it has started in a *special* configuration.

But now let's play a different game. First, shuffle the pack in a random way, then *look at* the first six cards and commit them to memory. Shuffle a little and look to see which other cards have ended up among the first six. At the start, there were none, then their number grew, as it did in the previous example, together with the entropy. But there is a crucial difference between this example and the previous one: at the beginning of this one, the cards were in a *random* configuration. It was *you* who declared them to be particular, by taking note of which cards were in the front half of the pack at the beginning of the game.

The same may be true for the entropy of the universe: perhaps it was in no particular configuration. Perhaps we are the ones who belong to a particular physical system with respect to which its state can be particular.

But why should there be such a physical system, in relation to which the initial configuration of the universe turns out to be special? Because in the vastness of the universe, there are innumerable physical systems, and they interact with each other in ways that are even more numerous. Among these, through the endless game of probabilities and huge numbers, there will surely